



Auto Grade Using Natural Language Processing





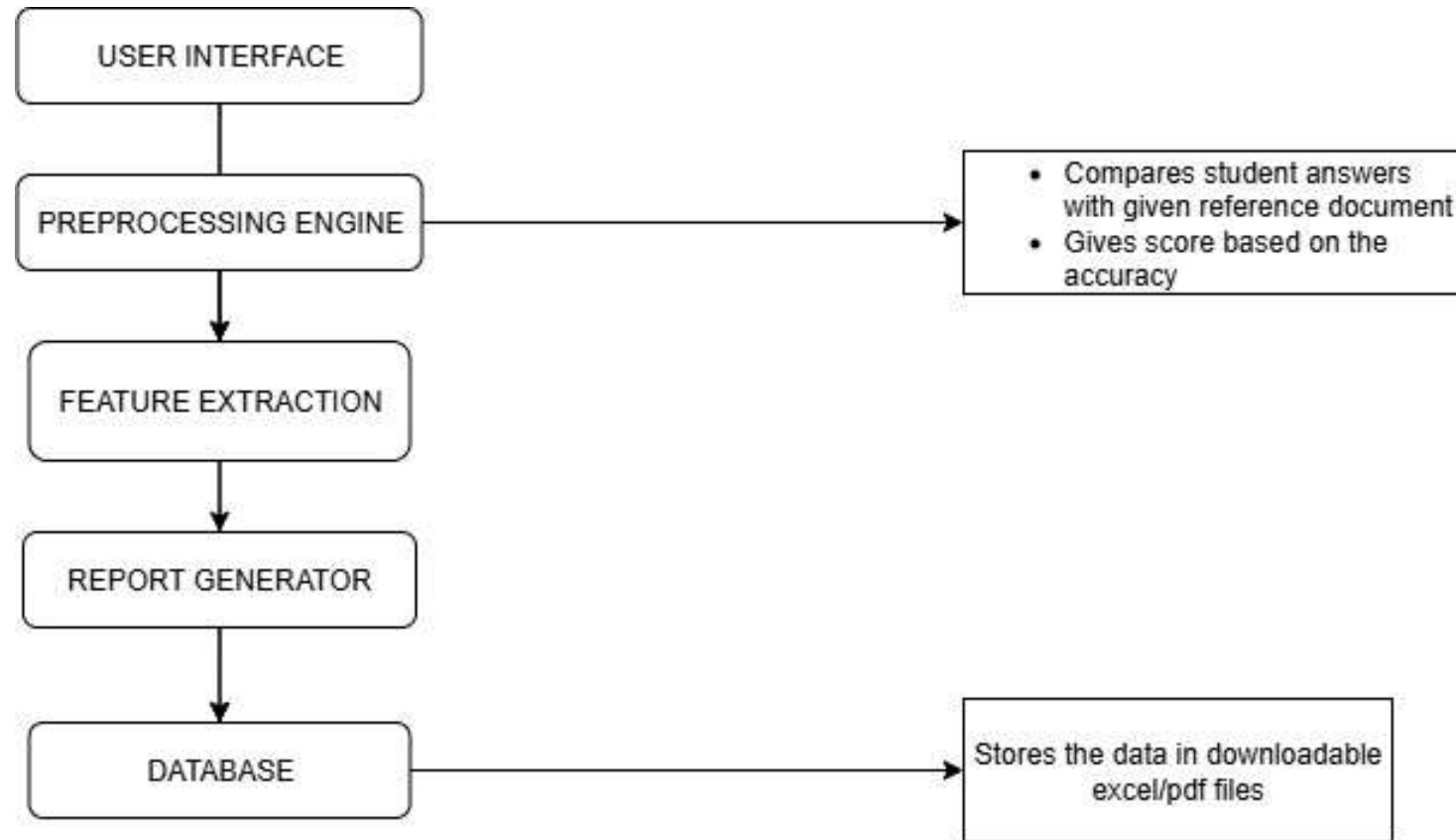
IMPLEMENTATION PLAN

- REQUIREMENT GATHERING
- SYSTEM DESIGN
- DEVELOPMENT
- TESTING
- DEPLOYMENT
- MAINTENANCE & FUTURE ENHANCEMENTS

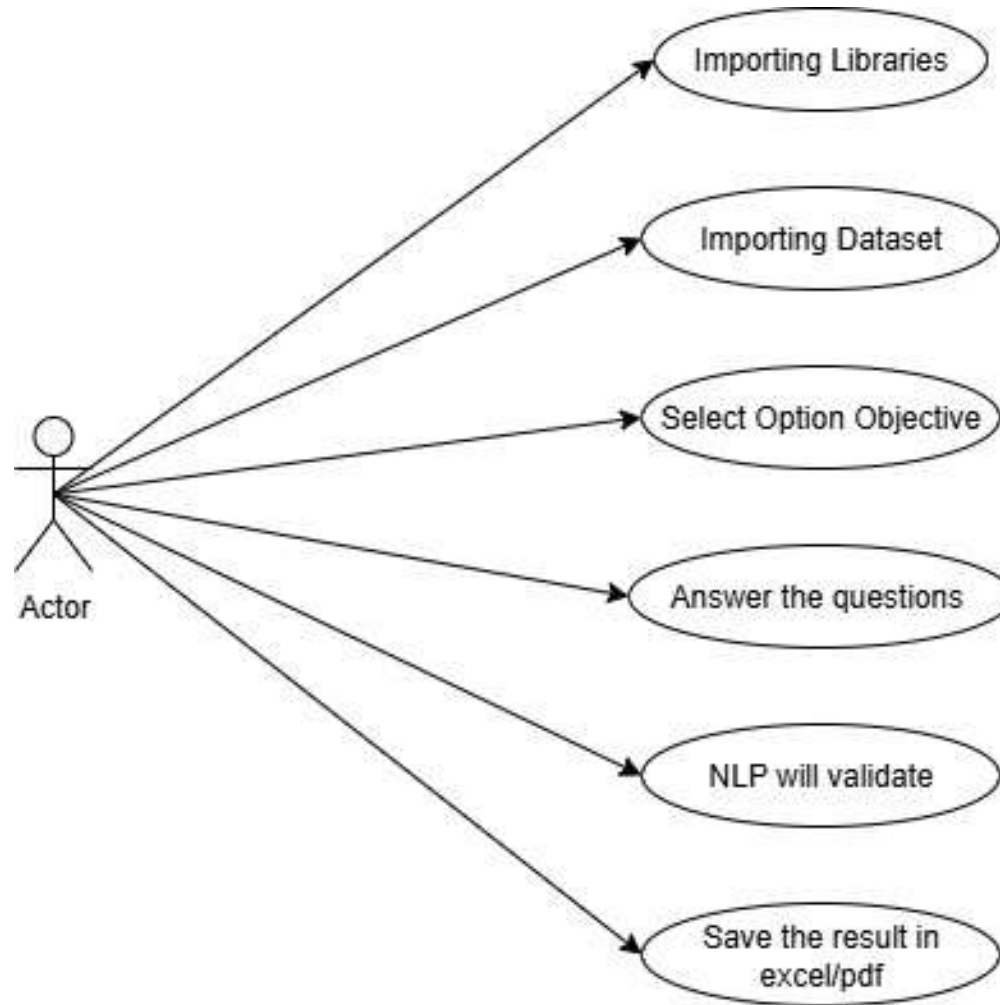
PHASE 1-REQUIREMENT GATHERING & ANALYSIS

- Identify Stakeholder End Users, Admins, Developers
- Define Supported File Types & Max Upload Size
- Set Performance Benchmarks (≤ 2 sec for short, ≤ 5 sec for long answers)
- Finalize Tech Stack
- Frontend: HTML, CSS, JavaScript (Bootstrap)
- Backend: Python
- Database: MySQL/PostgreSQL (production)
- Storage: Local in development, AWS S3 in production

PHASE 2- SYSTEM DESIGN - LOGICAL DESIGN



PHASE 2- SYSTEM DESIGN - UML DIAGRAM





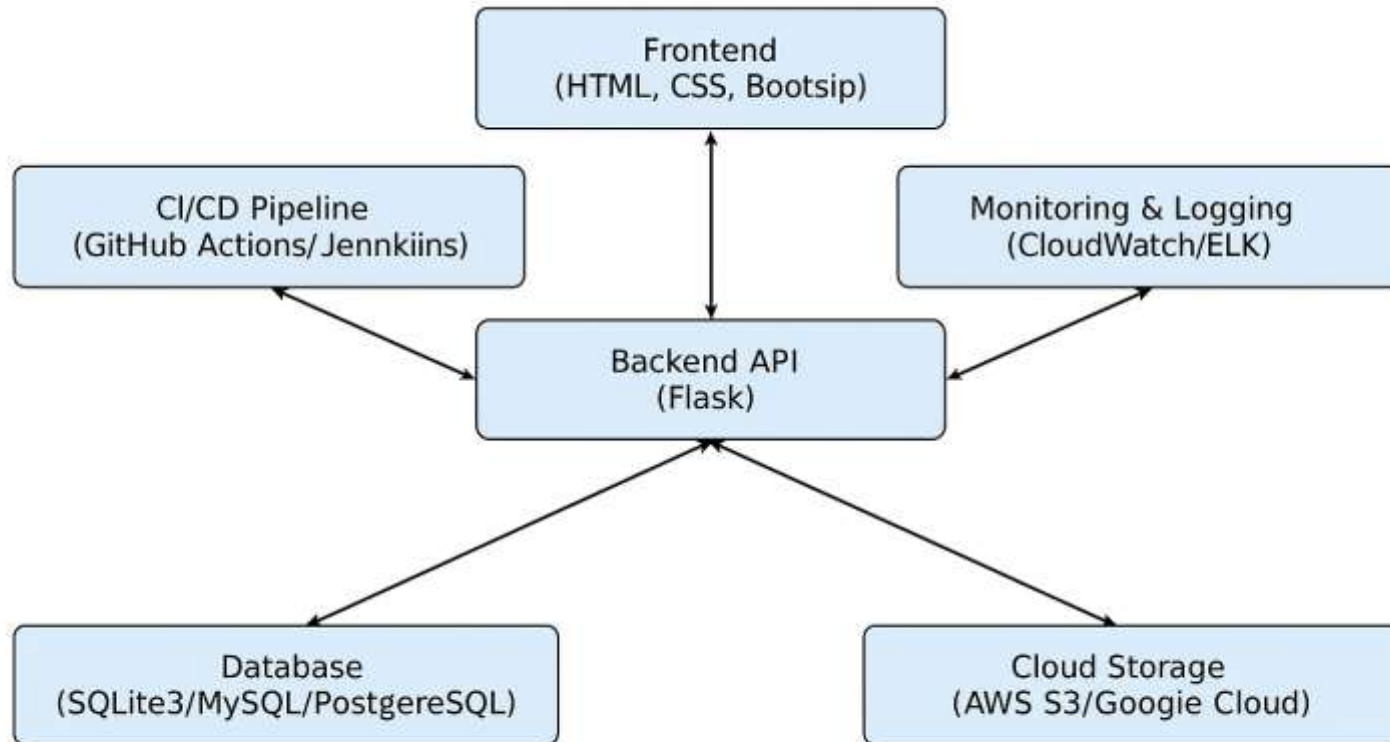
PHASE 3- DEPLOYMENT

- Answer Submission Initiation
- Metadata Extraction & Preliminary Check
- Text Preprocessing
- Feature Extraction
- Similarity Analysis
- Scoring Logic.
- Feedback Generation
- Result Storage & Metadata Update

PHASE 4- TESTING

- **Unit Testing:** Preprocessing, TF-IDF, BERT/SBERT, Similarity modules.
- **Integration Testing:** End-to-end grading flow (submission → scoring → storage).
- **Performance Testing:** 500+ concurrent submissions, ≤ 3 sec grading time.
- **Accuracy Monitoring:** Track grading accuracy, false positives/negatives, target $\geq 85\%$ match with human scoring.

PHASE 5- DEPLOYMENT



PHASE 6- MAINTENANCE AND FUTURE ENHANCEMENTS

- Monitor system performance & collect user feedback.
- Add support for more file types (e.g., audio, 3D models).
- Integrate blockchain for secure grading logs.
- Upgrade AI models (BERT → GPT-based) for higher accuracy.

THANK YOU

